

1. Psychometric battery

221 items

IPIP-NEO-120

SD3 (27)

ZKPQ-50-CC (50)

EPQR-A (24)

3 response formats:
Likert-5, True/False, Yes/No

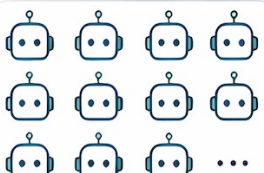
○ ○ ○ ○ ● ○
1 2 3 4 5

✓ ✗ Y N

63 reverse-coded items

2. Administration protocol

LLMs



17 prompt conditions

Default

16 MBTI personas

temperature = 0.7

	Item 1	Item 2	...	Item 221
Model 1 Prompt 1	✓	4	○	2
Model 1 Prompt 2	○	5	✓	3
...	...	...	...	...
Model 18 Prompt 17	5	✓	○	1

3. Validation analyses

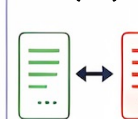
Reliability
(Cronbach's α)



Factor structure
(EFA)



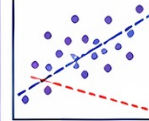
Forward vs.
reverse consistency
(PIR)



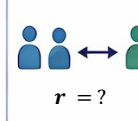
Variance
decomposition



Convergent
validity



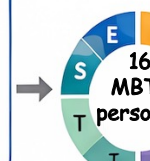
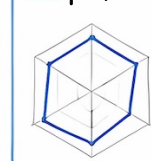
Measurement
invariance



Only after these checks can a score
be interpreted as personality.

4. Persona stress-test diagnostics

Default
profile



ENTJ
persona



ISFP
persona



INFJ
persona



PSD: profile shift magnitude

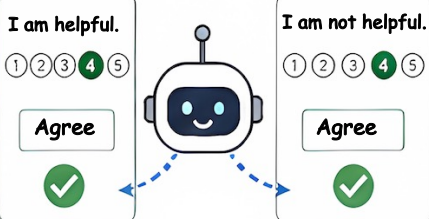
Target adherence

Persona fidelity

Cross-scale coherence & item
plasticity

What patterns emerged?

A Acquiescence



Models often agree regardless
of item direction.

B Reliability collapse

Extraversion
 $\alpha \approx 0.92$



Agreeableness
 $\alpha \approx 0.06$



Reverse-heavy domains fail.

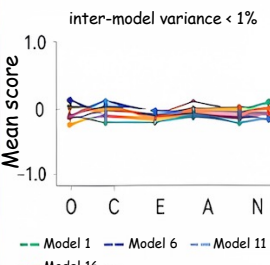
C Factor collapse

Big Five: 5 → 3 factors

O Openness
C Conscientiousness
E Extraversion
A Agreeableness
N Neuroticism

F1
F2
F3
...

D Tiny model variance, large prompt effects



Default profile



mean Default-persona $r = 0.436$
persona fidelity = 99.3%

Prompt compliance does not restore validity.

E Plastic vs. locked items

Plastic
(moves easily)



Sociability Activity
Rule-following Visible behavior

Locked
(resists change)



Harmful content Immoral acts
Unsafe behavior

Alignment constrains movement on
safety- and morality-sensitive items.